

Chapter 2: Molecular Genetics & DNA Replication

2.1 The DNA Double Helix Architecture

Antiparallel Complementary Strands

James Watson and Francis Crick, utilizing Rosalind Franklin's Photo 51 X-ray diffraction data, elucidated the antiparallel double-helical structure of deoxyribonucleic acid (DNA). The sugar-phosphate backbones run in opposite directions (5' to 3' vs 3' to 5'), stabilized by interior hydrogen bonds between complementary nitrogenous bases: Adenine pairs with Thymine (2 H-bonds) and Guanine pairs with Cytosine (3 H-bonds).

2.2 Semiconservative Replication & The Replication Fork

Leading vs Lagging Strands and Okazaki Fragments

Meselson and Stahl demonstrated that DNA replication is semi-conservative: each newly synthesized DNA double-helix contains one intact template strand from the parent molecule and one newly synthesized strand.

Enzymatic machinery: - Helicase unwinds the double helix at replication origins. - Topoisomerase relieves supercoiling tension ahead of the fork. - RNA Primase deposits short complementary RNA primers. - DNA Polymerase III synthesizes DNA strictly in the 5' to 3' direction. Because the strands are antiparallel, the leading strand is synthesized continuously, whereas the lagging strand must be synthesized discontinuously in short segments called Okazaki fragments, subsequently joined by DNA Ligase.

Key Takeaways:

- DNA Polymerase can only extend strands in the 5' to 3' direction, requiring a pre-existing 3' -OH group.
- Okazaki fragments on the lagging strand are sealed by phosphodiester bonds formed by DNA ligase.
- Proofreading by 3' to 5' exonuclease activity ensures astronomical replication fidelity (< 1 error per 10¹⁰ bases).